23AD653 MINI PROJECT - IV: NATURAL LANGUAGE PROCESSING

MODULE 1

Module Objective

To introduce fundamental text processing techniques used in Natural Language Processing by exploring raw text documents and performing essential pre-processing operations. This module aims to enable learners to load and analyze text data, apply sentence and word tokenization, and validate the script of the text to ensure language consistency. By the end of this module, learners will be able to convert unstructured raw text into a clean, structured, and validated form suitable for further linguistic and semantic analysis in subsequent NLP modules.

1. 1. Explore Basic Text Processing operation on text document

Step 1: Problem Understanding & Requirement Analysis

Step 2: Text Document Collection

Step 3: Text Document Loading

Step 4: Basic Text Exploration

Step 5: Text Cleaning (Basic Level)

1. 2. Perform Text Pre-processing - I (Tokenization, Script validation)

Step 6: Sentence Tokenization

Step 7: Word Tokenization

Step 8: Script Validation

Step 9: Handling Mixed-Script Text

Step 10: Error Handling & Edge Case Testing

Step 11: Result Visualization

Step 12: Documentation & Conclusion

Step 1: Problem Understanding & Requirement Analysis

- Understand the importance of text preprocessing in NLP
- Identify input text formats (TXT, paragraph, document)
- Define expected output (tokens, validated text)
- Study challenges in raw text (noise, symbols, mixed scripts)

Problem statement:

Medical reports often contain complex terminology and technical language that are difficult for non-medical users to understand. There is a need for an NLP-based system that can analyze user-entered medical report text and generate clear, simplified explanations.

The objective of this project is to build a chatbot-based system that processes medical report content using NLP techniques and provides understandable interpretations without performing medical diagnosis.

Input–output specification:

Input: Medical report content entered manually as unstructured text through a chatbot interface. The text may contain medical terms, test values, numbers, and observations.

Output: A cleaned and analyzed version of the text along with a simplified, easy-to-understand explanation of the medical terms and findings, presented in chatbot format.

Step 2: Text Document Collection

- Collect sample English text documents
- Include text with:
  - Punctuation
  - Numbers
  - Special characters
  - Mixed scripts (English + other languages)

- Mtsamples.csv (Kaggle.com)

  Features: description, Medical_speciality, smaple_name, Transcription, Keywords.

  No.of Rows: 5000

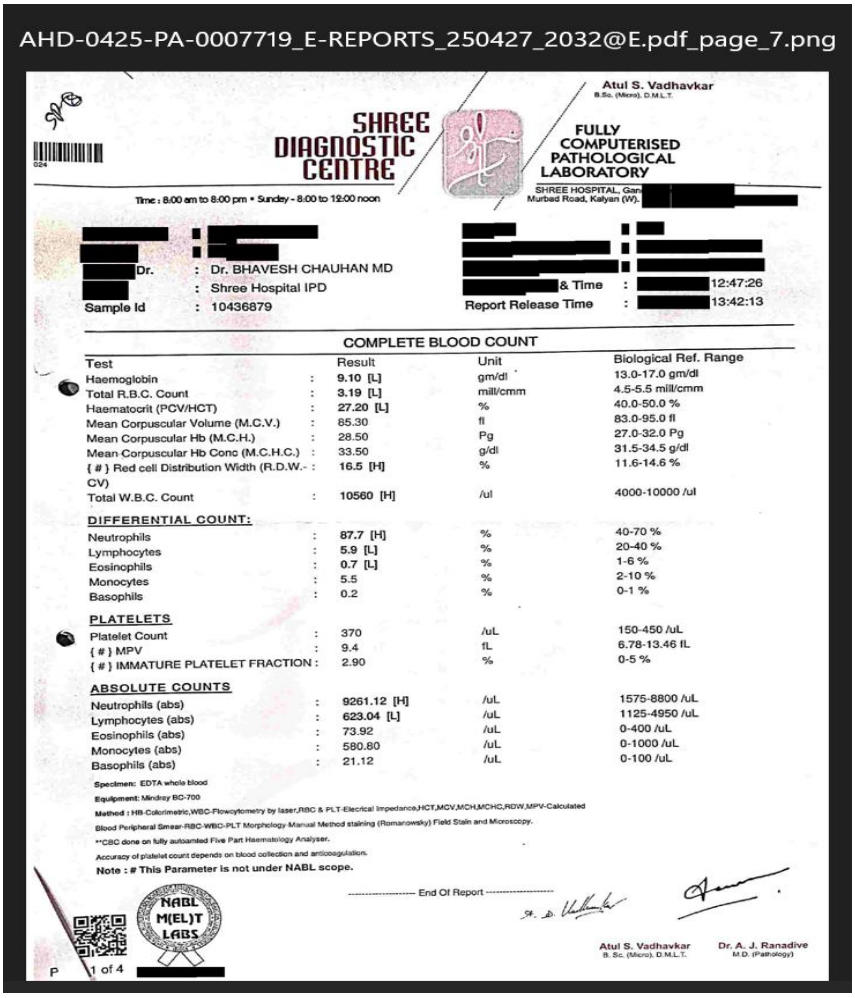
B2 | A 23-year-old white female presents with complaint of allergies.

	description	medical_specialty	sample_name	transcription	keywords
0	A 23-year-old white female	Allergy / Immunology	Allergic Rhinitis	SUBJECTIVE:, This 23-year-old white fer	allergy / immunology, allergic rhinitis, allergies, asthma, nasal sprays, rhinitis, nasal, ery
1	Consult for laparoscopic gas	Bariatrics	Laparoscopic Gastr	PAST MEDICAL HISTORY:, He has difficu	bariatrics, laparoscopic gastric bypass, weight loss programs, gastric bypass, atkin's die
2	Consult for laparoscopic gas	Bariatrics	Laparoscopic Gastr	HISTORY OF PRESENT ILLNESS: , I have s	bariatrics, laparoscopic gastric bypass, heart attacks, body weight, pulmonary embolisn
3	2-D M-Mode. Doppler.	Cardiovascular / Pulmo	2-D Echocardiograr	2-D M-MODE: , ,1. Left atrial enlargeme	cardiovascular / pulmonary, 2-d m-mode, doppler, aortic valve, atrial enlargement, dia
4	2-D Echocardiogram	Cardiovascular / Pulmo	2-D Echocardiograr	1. The left ventricular cavity size and wa	cardiovascular / pulmonary, 2-d, doppler, echocardiogram, annular, aortic root, aortic
5	Morbid obesity. Laparosco	Bariatrics	Laparoscopic Gastr	PREOPERATIVE DIAGNOSIS: , Morbid ob	bariatrics, gastric bypass, eea anastomosis, roux-en-y, antegastric, antecolic, morbid ob
6	Liposuction of the	Bariatrics	Liposuction	PREOPERATIVE DIAGNOSES:,1. Deformi	bariatrics, breast reconstruction, excess, lma anesthesia, lipodystrophy, liposuction, ab
7	2-D Echocardiogram	Cardiovascular / Pulmo	2-D Echocardiograr	2-D ECHOCARDIOGRAM,Multiple views	cardiovascular / pulmonary, 2-d echocardiogram, cardiac function, doppler, echocardic
8	Suction-assisted lipectomy -	Bariatrics	Lipectomy - Abdom	PREOPERATIVE DIAGNOSIS: , Lipodystro	bariatrics, lipodystrophy, abd pads, suction-assisted lipectomy, abdomen, aspirate, lipe
9	Echocardiogram and Doppl	Cardiovascular / Pulmo	2-D Echocardiograr	DESCRIPTION:,1. Normal cardiac chamb	cardiovascular / pulmonary, ejection fraction, lv systolic function, cardiac chambers, re
10	Morbid obesity. Laparosco	Bariatrics	Laparoscopic Gastr	PREOPERATIVE DIAGNOSIS: , Morbid ob	bariatrics, morbid obesity, roux-en-y, gastric bypass, antecolic, antegastric, anastomosis
11	Normal left ventricle, mode	Cardiovascular / Pulmo	2-D Doppler	2-D STUDY,1. Mild aortic stenosis, widel	cardiovascular / pulmonary, 2-d study, doppler, tricuspid regurgitation, heart pressures
12	Cerebral Angiogram - moya	Neurology	Moyamoya Disease	CC:, Confusion and slurred speech.,HX , (primarily obtained from boyfriend): This 31 y/o RHF experienced a "flu-like illness 6-8	
13	Patient presented to the ba	Bariatrics	Gastric Bypass Disc	PAST MEDICAL HISTORY:, Significant fo	bariatrics, weight watchers, roux en y, atkins, medifast, meridia, south beach, cabbage,
14	Surgical removal of comple	Dentistry	Bony Impacted Tee	PREOPERATIVE DIAGNOSIS:, Completel	dentistry, intraoral, bony impacted teeth, throat pack, buccal aspect, saline solution, gu
15	Preoperative visit for weigh	Bariatrics	Laparoscopic Gastr	HISTORY OF PRESENT ILLNESS: ,I have se	bariatrics, laparoscopic gastric banding, pulmonary embolism, lap banding, potential co
16	Neck exploration; tracheost	Cardiovascular / Pulmo	Tracheostomy	PREOPERATIVE DIAGNOSES,Airway obst	cardiovascular / pulmonary, airway, laryngology, shiley, alteration of voice, bronchi, br
17	Patient status post lap band	Bariatrics	Lap Band Adjustme	REASON FOR VISIT:, Lap band adjustme	bariatrics, lap band adjustment, lap band placement, lap band,
18	Fertile male with completed	Urology	Vasectomy - 4	PROCEDURE: , Elective male sterilization	urology, sterilization, vas, fertile male, bilateral vasectomy, vasectomy, cauterized,
19	The patient is a 17-year-old	General Medicine	Airway Compromis	HISTORY OF PRESENT ILLNESS:, The pat	general medicine, diabetes, hypertension, asthma, cholecystectomy, fishbone, foreign
20	Whole body radionuclide b	Urology	Whole Body Radio	INDICATION:, Prostate Cancer.,TECHNI	urology, prostate cancer, technetium, whole body, urinary retention, bone scan, radion
21	Patient discharged after lap	Bariatrics	Gastric Bypass Sum	FINAL DIAGNOSES,1. Morbid obesity, st	bariatrics, medifast, laparoscopic roux-en-y gastric bypass, roux-en-y, bariatric clear liq
22	Normal vasectomy	Urology	Vasectomy - 1	DESCRIPTION:, The patient was placed	urology, vasectomy, allis clamp, catgut, hemoclips, iris scissors, scrotal, scrotal supporte
23	Voluntary sterility. Bilateral	Urology	Vasectomy	PREOPERATIVE DIAGNOSIS: , Voluntary	urology, hemiscrotum, bilateral vasectomy, voluntary sterility, vas deferens, vasectomy
24	Blood in urine - Transitional	Urology	Urology Consut - 1	CHIEF COMPLAINT:,	
25	Normal vasectomy	Urology	Vasectomy - 2	DIAGNOSIS:, Desires vasectomy.,NAME	urology, scrotal incision, right vas, bleeding, anesthesia, vasectomy
26	Hispanic male patient was a	Urology	Urology Discharge	PROCEDURES:, Cystourethroscopy and t	urology, tur, bun, cystourethroscopy, difficulty voiding, bladder neck obstruction, creati
27	Umbilical hernia repair tem	Urology	Umbilical Hernia Re	PREOPERATIVE DIAGNOSIS: , Umbilical h	urology, marcaine, steri-strips, mattress sutures, umbilical hernia, repair, umbilical, her
28	Vasectomy 10 years ago, fai	Urology	Urology Letter	Sample Address,RE: Sample Patient,Wil	urology, letter, urology letter, azoospermic, cryopreservation, specimen harvest, sperm

Fig 1.1: Raw Medical Dataset in CSV

- Medical_reports Image dataset(Kaggle.com):

  No.of Images: 426 (PNG format)



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SHREE DIAGNOSTIC CENTRE

FULLY COMPUTERISED PATHOLOGICAL LABORATORY

SHREE HOSPITAL, Gan[redacted]
Murbad Road, Kalyan (W). [redacted]

Time : 8.00 am to 8.00 pm • Sunday - 8.00 to 12.00 noon

[redacted] : [redacted]
[redacted] : [redacted]
[redacted]Dr. : Dr. BHAVESH CHAUHAN MD
[redacted] : Shree Hospital IPD
Sample Id : 10436879

[redacted] : [redacted]
[redacted] : [redacted]
[redacted] : [redacted]
[redacted]& Time : [redacted]12:47:26
Report Release Time : [redacted]13:42:13

COMPLETE BLOOD COUNT

Test		Result	Unit	Biological Ref. Range
Haemoglobin	:	9.10 [L]	gm/dl	13.0-17.0 gm/dl
Total R.B.C. Count	:	3.19 [L]	mill/cmm	4.5-5.5 mill/cmm
Haematocrit (PCV/HCT)	:	27.20 [L]	%	40.0-50.0 %
Mean Corpuscular Volume (M.C.V.)	:	85.30	fl	83.0-95.0 fl
Mean Corpuscular Hb (M.C.H.)	:	28.50	Pg	27.0-32.0 Pg
Mean Corpuscular Hb Conc (M.C.H.C.)	:	33.50	g/dl	31.5-34.5 g/dl
{ # } Red cell Distribution Width (R.D.W.-CV)	:	16.5 [H]	%	11.6-14.6 %
Total W.B.C. Count	:	10560 [H]	/ul	4000-10000 /ul
DIFFERENTIAL COUNT:				
Neutrophils	:	87.7 [H]	%	40-70 %
Lymphocytes	:	5.9 [L]	%	20-40 %
Eosinophils	:	0.7 [L]	%	1-6 %
Monocytes	:	5.5	%	2-10 %
Basophils	:	0.2	%	0-1 %
PLATELETS				
Platelet Count	:	370	/uL	150-450 /uL
{ # } MPV	:	9.4	fL	6.78-13.46 fL
{ # } IMMATURE PLATELET FRACTION	:	2.90	%	0-5 %
ABSOLUTE COUNTS				
Neutrophils (abs)	:	9261.12 [H]	/uL	1575-8800 /uL
Lymphocytes (abs)	:	623.04 [L]	/uL	1125-4950 /uL
Eosinophils (abs)	:	73.92	/uL	0-400 /uL
Monocytes (abs)	:	580.80	/uL	0-1000 /uL
Basophils (abs)	:	21.12	/uL	0-100 /uL

Specimen: EDTA whole blood
Equipment: Mindray BC-700
Method : HB-Colorimetric,WBC-Flowcytometry by laser,RBC & PLT-Elecrical Impedance,HCT,MCV,MCH,MCHC,RDW,MPV-Calculated
Blood Peripheral Smear-RBC-WBC-PLT Morphology-Manual Method staining (Romanowsky) Field Stain and Microscopy.
**CBC done on fully autoamted Five Part Haematology Analyser.
Accuracy of platelet count depends on blood collection and anticoagulation.
Note : # This Parameter is not under NABL scope.

---------- End Of Report ----------

NABL M(EL)T LABS

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P 1 of 4 [redacted]

Fig 1.2: Sample Medical reports scanned copy

Step 3: Text Document Loading

- Read text from file or user input
- Handle encoding issues (UTF-8)
- Display raw text for verification

CODE:

```
df = pd.read_csv(r"dataset/mtsamples.csv", encoding='utf-8')
print("Dataset Loaded Successfully")
print("Shape:", df.shape)
print("\nSample Records:")
print(df.head())
print("\nRaw Text Sample:")
print(df.iloc[0]['transcription'][:500])
```

OUTPUT:

```
Dataset Loaded Successfully
Shape: (4999, 6)

Sample Records:
   Unnamed: 0                                       description  \
0           0   A 23-year-old white female presents with comp...
1           1           Consult for laparoscopic gastric bypass.
2           2           Consult for laparoscopic gastric bypass.
3           3                              2-D M-Mode. Doppler.
4           4                                  2-D Echocardiogram

              medical_specialty                               sample_name  \
0         Allergy / Immunology                          Allergic Rhinitis
1                   Bariatrics   Laparoscopic Gastric Bypass Consult - 2
2                   Bariatrics   Laparoscopic Gastric Bypass Consult - 1
3   Cardiovascular / Pulmonary                    2-D Echocardiogram - 1
4   Cardiovascular / Pulmonary                    2-D Echocardiogram - 2

                                     transcription  \
0  SUBJECTIVE:,  This 23-year-old white female pr...
1  PAST MEDICAL HISTORY:, He has difficulty climb...
2  HISTORY OF PRESENT ILLNESS: , I have seen ABC ...
3  2-D M-MODE: , ,1.  Left atrial enlargement wit...
4  1.  The left ventricular cavity size and wall ...

                                          keywords
0  allergy / immunology, allergic rhinitis, aller...
1  bariatrics, laparoscopic gastric bypass, weigh...
2  bariatrics, laparoscopic gastric bypass, heart...
3  cardiovascular / pulmonary, 2-d m-mode, dopple...
4  cardiovascular / pulmonary, 2-d, doppler, echo...

Raw Text Sample:
SUBJECTIVE:,  This 23-year-old white female presents with complaint of allergies.  She used to have allergies when she lived in Seattle but she thinks t
hey are worse here.  In the past, she has tried Claritin, and Zyrtec.  Both worked for short time but then seemed to lose effectiveness.  She has used A
llegra also.  She used that last summer and she began using it again two weeks ago.  It does not appear to be working very well.  She has used over-the-
counter sprays but no prescription nasal spr
```

Fig 1.3: Sample of Raw Data

Step 4: Basic Text Exploration

- Count:
  - Number of characters
  - Number of words
  - Number of sentences
- Identify punctuation and special characters
- Observe whitespace patterns

CODE:

```
for text in df['transcription'].astype(str):
        total_characters += len(text)
        total_words += len(word_tokenize(text))
        total_sentences += len(sent_tokenize(text))
        all_special_chars.extend(re.findall(r'[^\w\s]', text))
unique_special_chars = set(all_special_chars)
print("Total Characters:", total_characters)
print("Total Words:", total_words)
print("Total Sentences:", total_sentences)
print("Total Special Characters:", len(all_special_chars))
print("Unique Special Characters:", unique_special_chars)
```

OUTPUT:

Total Characters: 10420528

Total Words: 1869908

Total Sentences: 99367

Total Special Characters: 336409

Unique Special Characters: {'+', '<', '>', '?', '=', '*', '%', '&', '“', '"', '$', '"', '@', '®', '°', ')', '·', ’’, ';', '#', ':', ']', '.', '/', ’”’, '!', '(', '-', '©', '–', ',', '['}

Step 5: Text Cleaning (Basic Level)

- Convert text to lowercase
- Remove extra whitespaces
- Remove or handle punctuation
- Remove unwanted symbols

CODE:

```
def lowercase(text):
        return text.lower()

def remove_punctuation(text):
        return re.sub(r'[^\w\s]', ' ', text)

def remove_numbers(text):
        return re.sub(r'\d+', '', text)

def remove_special_characters(text):
        return re.sub(r'[^a-zA-Z\s]', ' ', text)

def remove_extra_whitespace(text):
        return re.sub(r'\s+', ' ', text).strip()

def remove_stopwords(text):
        tokens = word_tokenize(text)
        filtered = [w for w in tokens if w not in STOP_WORDS]
                return ' '.join(filtered)
```

Step 6: Sentence Tokenization

- Split text into sentences
- Handle sentence delimiters (. ? !)
- Display sentence list

CODE:

```
df['sentence_list'] = df['transcription'].apply(sent_tokenize)
print("\nSentence Token Sample:\n")
print(df['sentence_list'].head())
```

OUTPUT:

```
Sentence Token Sample:

0    [SUBJECTIVE:,  This 23-year-old white female p...
1    [PAST MEDICAL HISTORY:, He has difficulty clim...
2    [HISTORY OF PRESENT ILLNESS: , I have seen ABC...
3    [2-D M-MODE: , ,1., Left atrial enlargement wi...
4    [1., The left ventricular cavity size and wall...
Name: sentence_list, dtype: object
```

Step 7: Word Tokenization

- Split sentences into words
- Handle:
  - Apostrophes
  - Hyphenated words
- Display word tokens

CODE:

```
df['word_list'] = df['transcription'].apply(word_tokenize)
print("\nWord Token Sample:\n")
print(df['word_list'].head())
```

OUTPUT:

```
Word Token Sample:

0    [SUBJECTIVE, :, ,, This, 23-year-old, white, f...
1    [PAST, MEDICAL, HISTORY, :, ,, He, has, diffic...
2    [HISTORY, OF, PRESENT, ILLNESS, :, ,, I, have,...
3    [2-D, M-MODE, :, ,, ,1, ., Left, atrial, enlar...
4    [1, ., The, left, ventricular, cavity, size, a...
Name: word_list, dtype: object
```

Step 8: Script Validation

- Identify script of each word or sentence
- Validate whether text belongs to:
  - English (Latin script)
  - Non-English or mixed script
- Flag invalid or mixed-script text

CODE:

```
df['clean_word_list'] = df['clean_text'].apply(word_tokenize)
def is_english(word):
        return bool(re.fullmatch(r'[A-Za-z]+', word))
df['script_validation'] = df['clean_word_list'].apply(
    lambda words: all(is_english(w) for w in words)
)
print("\nScript Validation Report (After Cleaning):\n")
print(df['script_validation'].head())
```

OUTPUT:

```
Script Validation Report (After Cleaning):

0    True
1    True
2    True
3    True
4    True
Name: script_validation, dtype: bool
```

Step 9: Handling Mixed-Script Text

- Separate English text from non-English text
- Remove or mark invalid script content
- Retain only valid English tokens

```
df['filtered_english_text'] = df['word_list'].apply(
    lambda words: " ".join([w for w in words if is_english(w)])
)
print("\nFiltered English Text Sample:\n")
print(df['filtered_english_text'].head())
```

OUTPUT:

```
Filtered English Text Sample:

0    SUBJECTIVE This white female presents with com...
1    PAST MEDICAL HISTORY He has difficulty climbin...
2    HISTORY OF PRESENT ILLNESS I have seen ABC tod...
3    Left atrial enlargement with left atrial diame...
4    The left ventricular cavity size and wall thic...
Name: filtered_english_text, dtype: object
```

Step 10: Error Handling & Edge Case Testing

- Handle:
  - Empty files
  - Numeric-only text
  - Special-character-only text
- Ensure robust execution

CODE:

```
def validate_text(text):
        if not str(text).strip():
                return "Empty Text"
        if re.fullmatch(r'\d+', str(text)):
                return "Numeric Only"
        if re.fullmatch(r'[^\w\s]+', str(text)):
                return "Special Characters Only"
        return "Valid Text"
df['edge_case_check'] = df['transcription'].apply(validate_text)
print("\nEdge Case Check Sample:\n")
print(df['edge_case_check'].head())
```

OUTPUT:

```
Edge Case Check Sample:

0    Valid Text
1    Valid Text
2    Valid Text
3    Valid Text
4    Valid Text
Name: edge_case_check, dtype: object
```

Step 11: Result Visualization

- Display:
  - Raw text vs cleaned text
  - Sentence tokens
  - Word tokens
  - Script validation output

CODE:

```
plt.figure()
plt.hist(df['word_count_raw'], bins=30)
plt.title("Word Count Distribution")
plt.xlabel("Words per Document")
plt.ylabel("Frequency")
plt.show()

plt.figure()
plt.hist(df['sentence_count_raw'], bins=30)
plt.title("Sentence Count Distribution")
plt.xlabel("Sentences per Document")
plt.ylabel("Frequency")
plt.show()
```

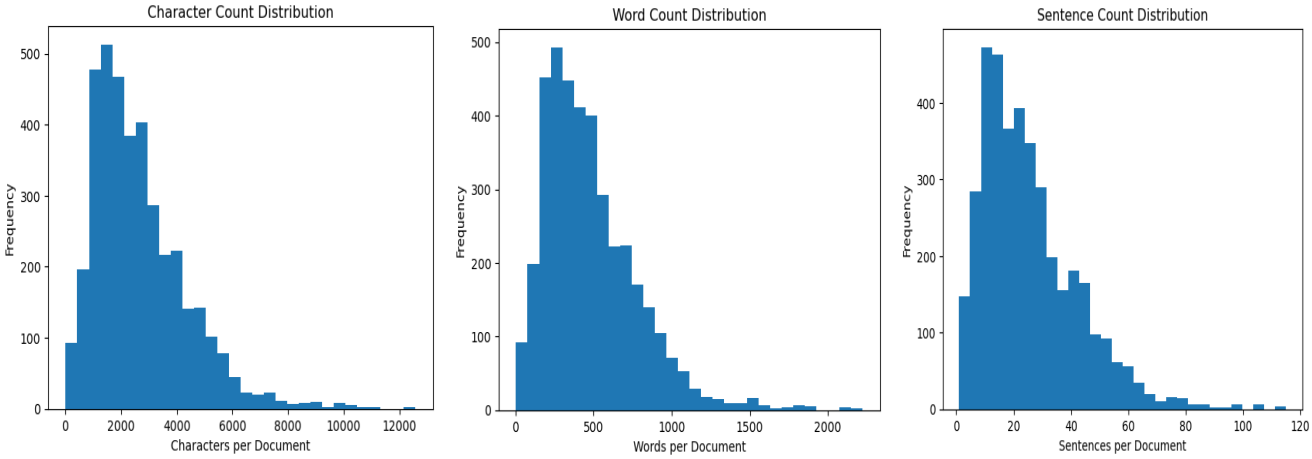

Fig 1.4: Visualization of dataset distribution

Step 12: Documentation & Conclusion

- Summarize steps performed

- Mention tools and techniques used
- Explain learning outcomes

MODULE 1 SUMMARY

1. Loaded dataset with UTF-8 encoding

2. Explored text statistics

3. Cleaned text

4. Applied sentence tokenization

5. Applied word tokenization

6. Performed script validation

7. Filtered English text

8. Handled edge cases

9. Generated visualizations

RESULT:

The text pre-processing module was successfully implemented by performing cleaning, sentence and word tokenization, script validation, mixed-script handling, and data visualization to prepare structured text for NLP tasks.